

# 3 · Experiments

## Datasets, baselines, and key results.

### Setup

QM9 (small organics, up to 9 atoms) and ZINC-250k (drug-like). 500 diffusion steps; classifier-free guidance,  $\lambda = 3$  on QM9,  $\lambda = 2$  on ZINC. Two tasks. Targeting: 100 property values  $\times$  10 molecules each, scored by MAE. Optimization (Jin et al.): corrupt for 100 steps, denoise toward the target, keep the best candidate above a Tanimoto threshold.

### Baselines

FreeGress and DiGress for targeting; JT-VAE, CG-VAE, GCPN for optimization. All operate on a fixed graph size. GrIDDD is the only model that can insert and delete atoms while sampling.

### Targeting accuracy

GrIDDD lowers MAE on both targeting benchmarks. QM9 dipole moment  $\mu$ : 0.66 vs 0.74 (FreeGress). ZINC molecular weight: 4.89 vs 8.96, roughly half the error of the strongest baseline, with no target size supplied to the model at inference.

### Optimization, OOD, ablation

On graph-property optimization, QED success reaches 45.1% against 9.4% for GCPN, and LogP improvement is 2.70. Out of distribution, where models are trained up to 14 atoms, GrIDDD still produces 35% valid molecules at size 15; DiGress collapses to near zero. Removing insertion and deletion takes QED success from 45.1% to 33.8%, so size control, not the diffusion backbone alone, is what produces the gains.

Table 2: Results of property targeting on the ZINC-250k dataset.

| Method        | LogP               |               | QED                |               | MW                 |              |
|---------------|--------------------|---------------|--------------------|---------------|--------------------|--------------|
|               | MAE ↓              | Val. ↑        | MAE ↓              | Val. ↑        | MAE ↓              | Val. ↑       |
| Unconditional | 1.52 ± 0.12        | 86.1 %        | 0.15 ± 0.01        | 86.1 %        | 74.16 ± 6.71       | 86.1 %       |
| DiGress       | 0.74 ± 0.08        | 74.6 %        | 0.15 ± 0.01        | 85.1 %        | 20.92 ± 2.90       | 40.4 %       |
| FreeGress     | <b>0.17 ± 0.01</b> | 84.9 %        | 0.04 ± 0.01        | 84.9 %        | 8.96 ± 1.93        | 79.7 %       |
| GrIDDD        | 0.19 ± 0.02        | <b>87.9 %</b> | <b>0.04 ± 0.00</b> | <b>87.2 %</b> | <b>4.89 ± 0.57</b> | <b>84.2%</b> |

Table 2 — property targeting on ZINC-250k.

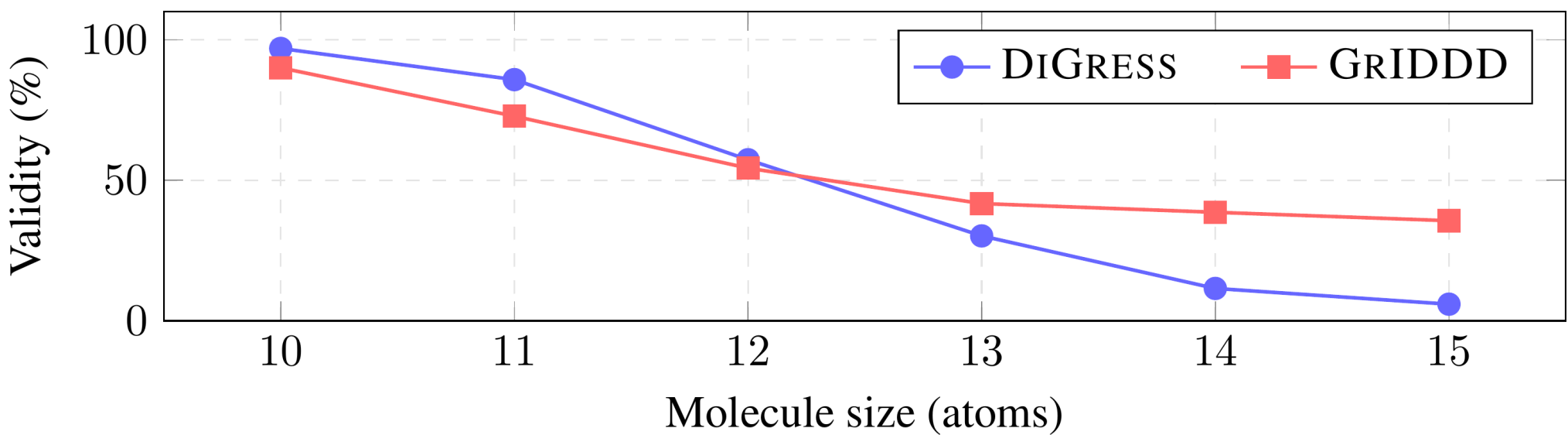


Figure 3: Validity results of out-of-distribution sampling on QM9.

Fig. 3 — out-of-distribution validity.

Table 3: Benchmark on graph property optimization.

| Method | LogP (sim ≥ 0.4)   |       | LogP (sim ≥ 0.6)   |       | QED (sim ≥ 0.4) |              | DRD2 (sim ≥ 0.4) |              |
|--------|--------------------|-------|--------------------|-------|-----------------|--------------|------------------|--------------|
|        | Improv. ↑          | Div.↑ | Improv. ↑          | Div.↑ | Succ.↑          | Div.↑        | Succ.↑           | Div.↑        |
| JT-VAE | 1.03 ± 1.39        | -     | 0.28 ± 0.79        | -     | 8.8 %           | -            | 3.4 %            | -            |
| CG-VAE | 0.61 ± 1.09        | -     | 0.25 ± 0.74        | -     | 4.8 %           | -            | -                | -            |
| GCPN   | 2.49 ± 1.30        | -     | 0.79 ± 0.63        | -     | 9.4 %           | 0.216        | 4.4 %            | <b>0.152</b> |
| GrIDDD | <b>2.70 ± 0.94</b> | 0.482 | <b>1.33 ± 0.61</b> | 0.280 | <b>45.1 %</b>   | <b>0.283</b> | <b>5.0 %</b>     | 0.121        |

Table 3 — graph-property optimization.